Name:_____ **Roll Number:**_____

Q1: Write the iterative formula for x_2^{k+1} and x_3^{k+1} for the Jacobi and Gauss - Seidel respectively, in a 3 x 3 system of equations, explicitly defining it in terms of the components of x^k and b. (2)

Q2: Use this system of linear equations and solve them using the (18)

$$
\begin{aligned}
2x_1 + 4x_2 - 2x_3 &= 2 \\
4x_1 + 9x_2 - 3x_3 &= 8 \\
-2x_1 - 3x_2 + 7x_3 &= 10
\end{aligned}
$$

a) Gaussian Elimination method.
b) LU Factorization method.
c) Cholesky Factorization method.
d) Compare the results obtained from these three methods and discuss the essential properties, applicability conditions, and computational drawbacks of these methods.
Solve on Answer Book